CN - IMP PYQ Analysis

COMPUTER NETWORKS | BE05000171 | Sem 5 | GTU

New Syllabus 2024-25 | P. Analyzed: 11 (W2020 – W2025)

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Chapter Reference

New Ch.	Covers
Ch. 1	Introduction, OSI/TCP-IP, Topologies, Transmission, Switching
Ch. 2	Data Link + Network Layer: Routing, IP, MAC, Error Detection
Ch. 3	Transport + Application Layer: TCP, UDP, DNS, HTTP, Sockets
Ch. 4	Wireless, IoT, Cloud Networking *(Zero PYQ - Blind Spot)*
Ch. 5	Network Security + Emerging Trends *(Zero PYQ - Blind Spot)*

M.IMP Questions - (High → Low Priority)

> **Priority Rule:** Freq ≥ 4 OR (Freq ≥ 3 AND Max Marks = 7) → M.IMP

1. CRC Error Detection with Polynomial Division and CRC Codeword Generation *(CRC)*

> **(Ch. 2)** | Freq: **9x** | Max Marks: **7** | `Theory: 3` `Num: 7`

2. Distance Vector Routing Algorithm with Router Table Updates and Network Example *(DVR)*

> **(Ch. 2)** | Freq: **9x** | Max Marks: **7** | `Theory: 9` `Num: 3` `Diag: 1`

3. HTTP Protocol: Persistent vs Non-Persistent Connections, Stateless Server, Request-Response Message Format *(HTTP)*

> **(Ch. 3)** | Freq: **9x** | Max Marks: **7** | `Theory: 9` `Def: 2`

4. DNS: Hierarchical System, Domain Name Resolution Process, Recursive and Iterative Query Types *(DNS)*

> **(Ch. 3)** | Freq: **8x** | Max Marks: **7** | `Theory: 8` `Diag: 4`

5. OSI Reference Model and TCP/IP Protocol Stack: Layer Architecture, Functions, and Comparison *(OSI/TCP-IP)*

> **(Ch. 1)** | Freq: **8x** | Max Marks: **7** | `Theory: 8` `Diag: 7`

6. Socket Programming System Calls: socket(), bind(), listen(), accept(), connect(), send(), recv() *(Socket API)*

> **(Ch. 3)** | Freq: **8x** | Max Marks: **7** | `Theory: 8` `Def: 2`

7. Go Back N and Selective Repeat Sliding Window Protocols: Mechanism, Differences, and Limitations *(GBN/SR)*

> **(Ch. 3)** | Freq: **7x** | Max Marks: **7** | `Theory: 7` `Diag: 7`

8. IP Subnetting: Classful and Classless Addressing, Subnet Mask Calculation, and VLSM with Examples *(Subnetting)*

> **(Ch. 2)** | Freq: **7x** | Max Marks: **7** | `Theory: 6` `Num: 5`

9. Link State Routing Algorithm with Dijkstra's Shortest Path Computation and Routing Table Construction *(LSR)*

> **(Ch. 2)** | Freq: **7x** | Max Marks: **7** | `Theory: 7` `Num: 2` `Diag: 4`

10. CSMA/CD and CSMA Variants: 1-Persistent, p-Persistent, Non-Persistent Working and Collision Detection *(CSMA)*

> **(Ch. 2)** | Freq: **6x** | Max Marks: **7** | `Theory: 6` `Diag: 4`

11. Email Protocols: SMTP Working, POP3 vs IMAP Comparison, and MIME Structure for Electronic Mail *(Email)*

> **(Ch. 3)** | Freq: **6x** | Max Marks: **7** | `Theory: 6`

12. Slotted ALOHA and Pure ALOHA: Working, Frame Transmission, Channel Utilization, and Efficiency *(ALOHA)*

> **(Ch. 2)** | Freq: **6x** | Max Marks: **7** | `Theory: 6` `Diag: 3`

13. TCP Segment Structure: Header Fields (SYN, FIN, ACK, URG, PSH, RST, Window, Checksum) and Field Significance *(TCP Segment)*

> **(Ch. 3)** | Freq: **6x** | Max Marks: **7** | `Theory: 6` `Diag: 4`

14. Connection-Oriented vs Connectionless Services: Comparison, Characteristics, and Use Cases *(CO vs CL)*

> **(Ch. 1)** | Freq: **6x** | Max Marks: **4** | `Theory: 6`

15. Unicast, Multicast, and Broadcast: Definitions, Packet Delivery Differences, and Network Implications *(UMB)*

> **(Ch. 2)** | Freq: **6x** | Max Marks: **4** | `Theory: 2` `Def: 4`

16. Virtual Circuit vs Datagram Networks: Working, Advantages, Disadvantages, and Comparison *(VC vs DG)*

(Ch. 2) | Freq: **6x** | Max Marks: **4** | `Theory: 6`

17. Count-to-Infinity Problem in DVR: Cause, Example, Split Horizon, and Poison Reverse Solutions *(Count-to-Inf)*

(Ch. 2) | Freq: **5x** | Max Marks: **7** | `Theory: 5`

18. Flow Control Protocols: Stop-n-Wait Working and Sliding Window Mechanism with Examples *(Flow Control)*

(Ch. 3) | Freq: **5x** | Max Marks: **7** | `Theory: 5` `Diag: 1`

19. IPv4 Address Classes A, B, C, D, E: Ranges, Default Subnet Masks, and Host Count Calculation *(IP Classes)*

(Ch. 2) | Freq: **5x** | Max Marks: **7** | `Theory: 4` `Num: 3`

20. Networking Device Functions: Repeater, Hub, Bridge, Switch, Router, and Gateway *(Net Devices)*

(Ch. 1) | Freq: **5x** | Max Marks: **7** | `Theory: 4` `Def: 1`

21. TCP Congestion Control: Slow Start, Congestion Avoidance, Fast Retransmit, and Fast Recovery Phases *(TCP CC)*

(Ch. 3) | Freq: **5x** | Max Marks: **7** | `Theory: 5` `Diag: 4`

22. TCP vs UDP: Detailed Feature, Service, Connection, and Use Case Comparison *(TCP vs UDP)*

(Ch. 3) | Freq: **5x** | Max Marks: **7** | `Theory: 5`

23. Types of Network Delay: Transmission, Propagation, Processing, and Queuing Delay with Formulas *(Net Delay)*

(Ch. 1) | Freq: **5x** | Max Marks: **4** | `Theory: 5` `Def: 2` `Num: 1`

24. Network Topologies: Star, Bus, Ring, Mesh, Tree Types with Advantages and Disadvantages *(Topologies)*

(Ch. 1) | Freq: **4x** | Max Marks: **7** | `Theory: 4` `Diag: 2` `Def: 1`

25. Transport Layer Multiplexing and Demultiplexing: Mechanism and Role of Port Numbers *(Mux/Demux)*

(Ch. 3) | Freq: **4x** | Max Marks: **7** | `Theory: 4` `Def: 1`

26. UDP (User Datagram Protocol): Header Fields, Services, and Advantages vs Disadvantages *(UDP)*

(Ch. 3) | Freq: **4x** | Max Marks: **7** | `Theory: 4` `Diag: 2`

27. Bit Rate and Baud Rate: Definitions, Relationship, and Differences in Data Transmission *(Bit/Baud Rate)*

(Ch. 1) | Freq: **4x** | Max Marks: **4** | `Theory: 3` `Def: 2`

28. Throughput in Computer Networks: Definition, Measurement, and Factors Affecting It *(Throughput)*

(Ch. 1) | Freq: **4x** | Max Marks: **3** | `Theory: 2` `Def: 4`

29. Circuit Switching vs Packet Switching: Working, Store-and-Forward, Comparison, and Tradeoffs *(CS vs PS)*

(Ch. 1) | Freq: **3x** | Max Marks: **7** | `Theory: 3`

30. Congestion Control vs Flow Control vs Error Control: Comparison and Differentiation *(CC vs FC vs EC)*

> **(Ch. 3)** | Freq: **3x** | Max Marks: **7** | `Theory: 3`

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31. IPv4 Datagram Header Format: All Fields, Functions, ToS, TTL, Protocol, and Fragmentation Fields *(IPv4 Header)*

> **(Ch. 2)** | Freq: **3x** | Max Marks: **7** | `Theory: 3` `Diag: 2`

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32. Port Numbers for Well-Known Application Layer Protocols: SMTP, HTTP, HTTPS, DNS, FTP, POP3, Telnet, SSH *(Well-Known Ports)*

> **(Ch. 3)** | Freq: **3x** | Max Marks: **7** | `Def: 3`

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33. Taking Turn MAC Protocols: Binary Countdown, Token Ring, and Reservation Protocol Working *(Taking Turn)* | (❌ Out_Of_Syllabus)

> **(Ch. 2)** | Freq: **3x** | Max Marks: **7** | `Theory: 3` `Diag: 1`

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34. TCP Flow Control and Error Control Mechanisms: Receive Window, ACK, Sequence Numbers, and Retransmission *(TCP FC/EC)*

> **(Ch. 3)** | Freq: **3x** | Max Marks: **7** | `Theory: 3`

IMP Questions - High → Low Priority

> **Priority Rule:** Freq 2–3 AND Max Marks ≥ 4 OR Freq ≥ 3 as 3-mark Definition → IMP

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1. Broadcast and Multicast Routing: Algorithms, Spanning Tree Approach, and Multicast Group Management *(B&M Routing)* | (❌ Out_Of_Syllabus)

(Ch. 2) | Freq: **3x** | Max Marks: **4** | `Theory: 3`

2. IPv4 vs IPv6: Header Format, Address Size, Key Features, and Transition Mechanism Comparison *(IPv4 vs IPv6)*

(Ch. 2) | Freq: **3x** | Max Marks: **4** | `Theory: 3`

3. NAT (Network Address Translation): Working, Private to Public IP Mapping, and IP Depletion Fix *(NAT)*

(Ch. 2) | Freq: **3x** | Max Marks: **4** | `Theory: 3`

4. Parity Check: Single-Bit Parity and Two-Dimensional Parity for Error Detection with Examples *(Parity Check)*

(Ch. 2) | Freq: **3x** | Max Marks: **4** | `Theory: 3` `Num: 1`

5. Encapsulation Process in Transport Layer: Headers, Segment Formation, and PDU Structure *(Encapsulation)*

(Ch. 3) | Freq: **3x** | Max Marks: **3** | `Theory: 2` `Def: 1`

6. Piggybacking in Flow and Error Control: Definition, Working Mechanism, and Advantage *(Piggybacking)*

(Ch. 3) | Freq: **3x** | Max Marks: **3** | `Theory: 2` `Def: 3`

7. DHCP Protocol: Dynamic IP Address Assignment, DORA Process, and Working *(DHCP)* | (❌ Out_Of_Syllabus)

(Ch. 1) | Freq: **2x** | Max Marks: **7** | `Theory: 2`

8. Ethernet Frame Structure and IEEE 802.3 Standard: Fields, MAC Addressing, and Frame Transmission *(Ethernet Frame)*

> **(Ch. 2)** | Freq: **2x** | Max Marks: **7** | `Theory: 2` `Diag: 2`

9. Leaky Bucket Algorithm for Traffic Shaping and Congestion Prevention *(Leaky Bucket)*

> **(Ch. 3)** | Freq: **2x** | Max Marks: **7** | `Theory: 2` `Diag: 2`

10. Reliable Data Transfer (RDT) Principles and RDT 2.0 Stop-and-Wait Protocol Design *(RDT)*

> **(Ch. 3)** | Freq: **2x** | Max Marks: **7** | `Theory: 2` `Diag: 1`

11. ARP (Address Resolution Protocol): IP-to-MAC Address Resolution, Request-Reply, and ARP Cache *(ARP)* | (❌ Out_Of_Syllabus)

> **(Ch. 2)** | Freq: **2x** | Max Marks: **4** | `Theory: 2`

12. Bit Stuffing for Frame Delimiting in Data Link Layer with Worked Example *(Bit Stuffing)*

> **(Ch. 2)** | Freq: **2x** | Max Marks: **4** | `Theory: 2` `Num: 2`

13. FQDN vs PQDN: Fully vs Partially Qualified Domain Names in DNS Context *(FQDN/PQDN)*

> **(Ch. 3)** | Freq: **2x** | Max Marks: **4** | `Theory: 2`

14. Physical Address vs Logical Address vs Port Address: Definitions, Differences, and Layer Mapping *(Addressing Levels)*

> **(Ch. 1)** | Freq: **2x** | Max Marks: **4** | `Theory: 2` `Def: 2`

15. Process-to-Process Delivery in Transport Layer: Role of Port Numbers and Socket Pairs *(P2P Delivery)*

> **(Ch. 3)** | Freq: **2x** | Max Marks: **4** | `Theory: 2`

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16. Route Aggregation and CIDR: Supernetting, Prefix Notation, and Routing Table Efficiency *(CIDR)*

> **(Ch. 2)** | Freq: **2x** | Max Marks: **4** | `Theory: 2`

New Syllabus Blind Spots - High → Low Priority

> **Zero PYQ Coverage - Study Anyway |** Priority ranked by exam likelihood. Topics exist in **Ch. 4 and Ch. 5** New syllabus only.

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1. Software Defined Networking (SDN): Control Plane vs Data Plane Separation, OpenFlow Protocol, and SDN Architecture

(Ch. 5) - **Why:** Highest-value emerging tech topic, near-certain 7-mark question in first exam cycle.

2. Firewall Types: Packet Filtering, Stateful Inspection, and Application-Level Gateway Working and Comparison

(Ch. 5) - **Why:** First and most fundamental security concept listed in Ch. 5; classic theory and diagram question with very high predictability.

3. 5G Network Architecture: Radio Access Network, Core Network, Network Slicing, and Key Differences from 4G

(Ch. 4) - **Why:** Prominently cited in syllabus rationale; GTU consistently targets hot-technology topics in first exams under new chapters.

4. VPN (Virtual Private Network): Tunneling Protocols, IPsec, SSL-VPN, and Site-to-Site vs Remote Access Types

(Ch. 5) - **Why:** Standard security chapter question globally; direct 4-mark or 7-mark theory question predictable from any CN exam pattern.

5. Wi-Fi Standards (IEEE 802.11 a/b/g/n/ac/ax): Speed, Frequency Band, Range, and Security Comparison

(Ch. 4) - **Why:** Explicitly listed in new Ch. 4; straightforward comparison table question is the most predictable exam format.

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6. Cryptography Basics: Symmetric vs Asymmetric Encryption, AES, RSA Algorithm, and Digital Signatures

(Ch. 5) - **Why:** Core prerequisite for all Ch. 5 security topics; CO-4 carries 22% marks weightage making security section mandatory.

7. MQTT and CoAP Protocols for IoT Communication: Architecture, Message Model, Differences, and Use Cases

(Ch. 4) - **Why:** Both protocols explicitly listed by name in new syllabus; definition and comparison question expected in any first exam under new code.

8. IDS and IPS (Intrusion Detection and Prevention Systems): Types, Working, and Signature vs Anomaly Detection

(Ch. 5) - **Why:** Directly listed in Ch. 5 content; standalone 3-to-4-mark theory question is the most likely format.

9. Cloud Networking: IaaS, PaaS, SaaS Models, CDN, and Elastic Load Balancing Concepts

(Ch. 4) - **Why:** GTU PBL activities explicitly include cloud infrastructure visit; cloud networking cited in course rationale as industry-relevant skill.

10. SSL/TLS Protocol and HTTPS: Handshake Process, Certificate Validation, and Secure Communication Mechanism

(Ch. 5) - **Why:** Bridges heavily-tested HTTP (Ch. 3 M.IMP) with Ch. 5 security topics; natural pairing question likely in the same paper.

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11. Virtualization and Container Networking: Hypervisors, VMs, Docker Architecture, and Container Networking Basics

(Ch. 4) - **Why:** Explicitly listed as "concept of container networking" in Ch. 4; DevOps relevance cited in course rationale.

12. Zero-Trust Network Security Model: Never-Trust-Always-Verify Principle, Micro-Segmentation, and Implementation

(Ch. 5) - **Why:** Specifically named in new Ch. 5 syllabus; buzzword concept likely asked as short essay or definition question.

13. AI-Driven Network Monitoring: Anomaly Detection, ML-Based Traffic Analysis, and Case Study Format

(Ch. 5) - **Why:** Listed as a case study topic in Ch. 5 content; case study format question is directly predictable from the syllabus wording.

14. Bluetooth Architecture: Piconet, Scatternet, Frequency Hopping Spread Spectrum, and Protocol Stack

(Ch. 4) - **Why:** Explicitly listed in Ch. 4 under wireless technologies; beginner-friendly definition and architecture question predictable.

15. Cybersecurity Attack Types and Mitigation: DDoS, Man-in-the-Middle, Phishing, and Botnet Attacks

(Ch. 5) - **Why:** DDoS and botnet were tested once in S2022 under old syllabus; new Ch. 5 cybersecurity practices section makes this a higher-probability repeat target.

Quick Stats Summary

Category	Count	Papers Analyzed
M.IMP Topics	34	11 (W2020 → W2025)
IMP Topics	16	11 (W2020 → W2025)
Blind Spots (Ch. 4+5)	15	N/A - New Syllabus Only
Highest Freq Topic	CRC / DVR / HTTP	9/11 papers each
Most Diag-Heavy	OSI/TCP-IP, GBN/SR	Diag in 7 papers each
Pure Def Topics	Well-Known Ports, UMB	Definition recall only

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